

Mahesh Sai Kalya

DATA ANALYST

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SUMMARY

- Data Analyst with 4+ years of experience delivering insights through SQL, Python, BI dashboards, and cloud-based analytics across healthcare and finance domains.
- Built automated ETL pipelines, predictive ML models, and analytics dashboards that improved reporting efficiency by 25–40% and accelerated data-driven decision making.
- Experienced in AWS, Snowflake, Databricks, Hadoop, and Spark, with a strong background in large-scale data processing and distributed computing.
- Skilled in HIPAA/GDPR-compliant data governance, EHR analytics (Epic, Cerner), and financial risk modeling, partnering with cross-functional teams to drive measurable business impact.

EDUCATION

- **Master of Science in Computer Technology** - Eastern Illinois University Charleston, IL, USA May 2025
- **Bachelor of Technology in Information Technology** - JNTU, Hyderabad, India July 2021

SKILLS

Programming Language:	Python, SQL, Scala, R
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, ggplot2
Visualization Tools:	Tableau, Power BI, Looker, Advanced Excel (Pivot Tables, Power Query)
Cloud Technologies:	AWS (EC2, S3, Redshift, Athena, Glue, Lambda, DynamoDB), Azure (Databricks), Snowflake
Database:	MySQL, PostgreSQL, MongoDB, SQL Server, Oracle
Other Technical:	Machine Learning, ETL (SSIS, Informatica MDM, Alteryx), Data Cleaning, Data Wrangling, A/B Testing, Hypothesis Testing, Forecasting, Clustering, Classification, Regression, ServiceNow, Git, GitHub, Jira, Data Mining, Data Visualization, Data Warehousing, Data Transformation, Query Optimization (CTEs, indexing, execution plans), Airflow, Power Automate, Data Modeling (Star/Snowflake Schema), REST API Integration, Time Series Forecasting, Data Quality Checks, Data Validation Frameworks, Feature Engineering, Distributed Data Processing, Bash/Shell Scripting, Docker, Snowflake Streams & Tasks, dbt (Data Build Tool), PySpark, Databricks, API-based Data Pipelines, Hadoop, Spark

WORK EXPERIENCE

HCA Healthcare, IL | Data Analyst

Oct 2024 - Present

- Engineered automated ETL pipelines using **AWS Glue, SQL Server, and Scala**, reducing data processing time by **35%** and improving dashboard refresh intervals for operational leaders.
- Designed and deployed **Power BI and Excel** dashboards integrated with **Snowflake**, improving reporting speed and analytical visibility by **25%** across clinical and operational teams.
- Developed patient churn and retention prediction models using TensorFlow and scikit-learn (with R used for supporting visualizations), improving identification of high-risk patients and enabling targeted intervention strategies.
- Performed large-scale **EDA** across millions of EHR records (demographics, treatments, appointments) to uncover key drivers of patient churn such as adherence gaps and satisfaction metrics.
- Extracted, transformed, and validated complex healthcare datasets using **advanced SQL, Pandas, and PySpark**, ensuring high data accuracy for analytics and machine learning workflows.
- Executed **A/B tests, ANOVA, and hypothesis testing** with SciPy to evaluate outreach strategies, directly supporting improved ROI for patient retention campaigns.
- Built reusable **feature engineering pipelines** to support predictive modeling, including creation of behavioral, engagement, and utilization-based features.
- Partnered with cross-functional stakeholders in **Agile sprints**, participating in UAT and ensuring analytics deliverables aligned with clinical, operational, and compliance requirements.

Infosys, India | Data Analyst

Nov 2020 – Aug 2023

- Built Tableau and SSRS dashboards for monitoring credit risk exposure, improving early risk detection accuracy by 27% across retail banking portfolios.
- Automated large-scale data ingestion workflows using Informatica MDM and Oracle SQL, reducing manual processing by 40% and enhancing data reliability.
- Conducted clustering and regression analysis using SAS and Alteryx to identify hidden customer behavior patterns and forecast delinquency with improved precision.
- Consolidated multi-source financial data using Azure Synapse and DynamoDB, improving traceability and speeding up access to historical market and transaction data.

- Developed custom Python-based validation and parsing scripts, improving data ingestion efficiency by 25% and reducing downstream reconciliation errors.
- Built statistical and financial models in R (base + tidyverse) and developed KPIs using Seaborn, enabling data-driven adjustments in credit policy.
- Applied SDLC and Agile methodologies to deliver analytics solutions aligned with evolving compliance and SLA requirements.
- Queried and aggregated datasets for 75,000+ clients using PostgreSQL and complex SQL joins to support investment risk and portfolio optimization.
- Improved overall data quality by 13% through Python automation, segmentation pipelines, and transformation logic across multiple financial data sources.
- Developed Power BI dashboards showcasing portfolio performance, risk indicators, and asset allocation trends for senior leadership.
- Built logistic regression models predicting portfolio underperformance to support proactive risk mitigation and investment decision-making.
- Implemented ServiceNow-based data quality rules and SQL validation checks, reducing recurring manual audit efforts by 30%.